Subject:	Database Development 171/181
Assessment:	Assignment 2
Total:	60 Marks

Introduction

The quest for better data management has led to several generations of databases that attempt to resolve critical data management shortcomings and to provide solutions to ever-evolving business needs. These database models use different approaches to represent data structures, their components, and relationships to support specific problem domains. The advent of the relational model, often considered the second-generation database model, marked a significant breakthrough in data management. Their conceptual simplicity set the stage for a genuine database revolution. A relational database management system (RDBMS) provides numerous functions that make the relational databases easier to understand and implement. There are many different commercial RDBMSs. In this module, we learn how to implement relational databases using one specific RDBMS called Microsoft™ Access.

Implementing a good database requires careful design and planning. A well-designed database facilitates data management and generates accurate and valuable information. The relational data model and Entity Relational Model (ERM) combines to provide the foundation for tightly structured database design. In the context of databases, Unified Modelling Language (UML) is a powerful tool for database architects and developers to plan and document their database designs effectively. It provides a visual representation of database structures, entities, attributes, and their relationships, making it easier to understand and communicate complex database designs.

Outline

With the above in mind, do some detailed research on databases and address the following topics:

1. Discuss the strengths and limitations of Microsoft Access databases. In your answer, identify specific use cases where Microsoft Access would be suitable and situations where it would not be appropriate.
2. Describe, with the aid of suitable examples, the UML notation for Entity Relationship Diagrams (ERD).
3. Explain the key principles and characteristics that define the relational database model.
4. Newer generations of databases emerged to address some very specific shortcomings of the hugely popular relational model. Describe these new and emerging data models, specifically discussing the problems they address.
5. With the emergence of many new types of databases, do you think that relational databases will still dominate the market share of database applications in the future? Use plausible arguments to back your answer.

Subject:	Database Development 171/181
Assessment:	Assignment 2
Total:	60 Marks

Mark Allocation

Content	
Strengths, Limitations and Use Case of Access databases	10
UML Notation for Entity Relationship Diagrams (ERDs)	10
Key Characteristics and Principles of Relational Databases	10
Emerging Data Models and Problems Addressed	15
The Future of Database Systems	10
Layout	2
References/Bibliography	3
Total:	**60**

Additional Information

- This is an individual assignment. All work must be done on your own.
- Submit your assignment electronically on Moodle (BC Connect) before the due date.
- All work must be original. Copying or reproducing someone's work will not be tolerated.
- Marks will be awarded for content, layout, logical flow of presentation and strengths of arguments.
- All writing must be correctly referenced and cited.
- There must be a Cover Page with your name, surname, and class on it.
- List all your sources (book titles, website addresses, journals, etc.) on the last page of your presentation under the headline "References".
- Belgium Campus makes use of software that can **scan for plagiarism** and penalties apply for any plagiarised work including getting a 0 for this assignment
- No mark will be awarded if the assignment is not uploaded via BC Connect.
- Late submissions will not be accepted; missing the deadline is an automatic **0**.